

CS388: Natural Language Processing

Lecture 12: ICL 2: Text rationales, Chain-of-thought



Slides credit: Greg Durrett



Administration

- ▶ Midterm coming up: 10/8
 - ▶ Topics up to and including 10/1 (not 10/6)
 - ▶ Similar format to quizzes
 - ▶ Content will skew towards new material not covered on quizzes
 - ▶ But quiz material is also fair game and some questions will address quiz material
 - ▶ You may bring a cheat sheet: one 8.5x11" sheet of paper (you can use both sides if you want), printed or hand-written.
 - ▶ if you bring a cheat sheet you'll have to turn it in with your exam



Recap: Zero-shot/Few-shot prompting

- ▶ Single unlabeled datapoint $\mathbf{x}$, want to predict label y

$\mathbf{x}$ = *The movie's acting could've been better, but the visuals and directing were top-notch.*

- ▶ Wrap $\mathbf{x}$ in a template we call a **verbalizer** $\mathbf{v}$

Review: *The movie's acting could've been better, but the visuals and directing were top-notch.*

Out of positive, negative, or neutral, this review is — **GPT-3** → neutral

- ▶ Need the right prompt (but there is a “plateau” of prompts that work)
- ▶ Few-shot: add one or more examples. Typically works better! Particularly with rich examples like we'll see today



Recap: Understanding ICL

- ▶ ICL can learn a strategy like ordinary least-squares (Akyurek et al., 2022)
- ▶ Is ICL doing gradient descent? Jury is out...



This Lecture

- ▶ Text rationales: text explanations of answers
- ▶ Chain-of-thought prompting (zero- and few-shot)
- ▶ Extensions
- ▶ Analysis of explanations

Text Rationales



Example from Vision

Laysan Albatross



Description: This is a large flying bird with black wings and a white belly.

Class Definition: The *Laysan Albatross* is a large seabird with a hooked yellow beak, black back and white belly.

Visual Explanation: This is a *Laysan Albatross* because this bird has a large wingspan, hooked yellow beak, and white belly.

Laysan Albatross

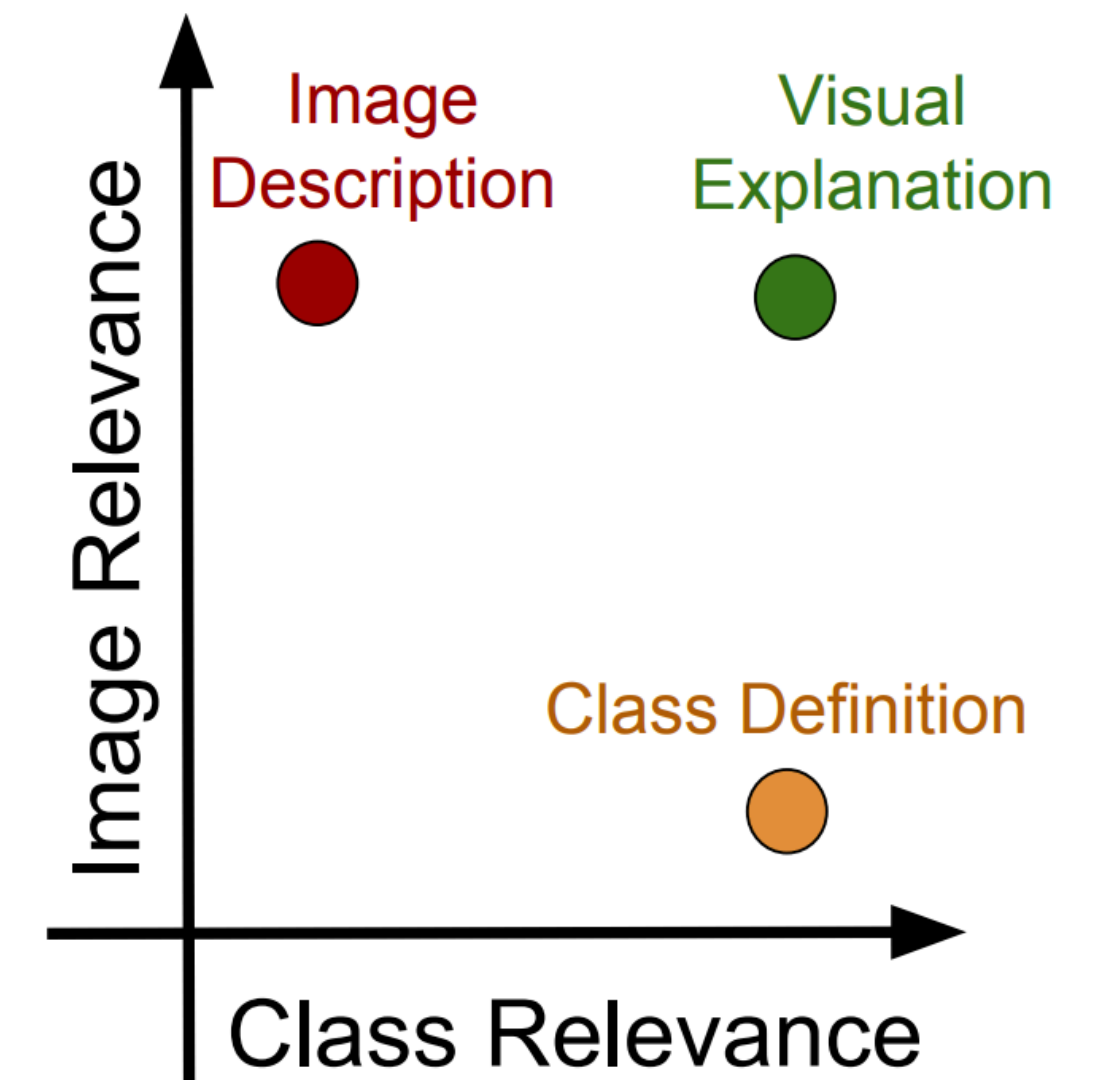


Description: This is a large bird with a white neck and a black back in the water.

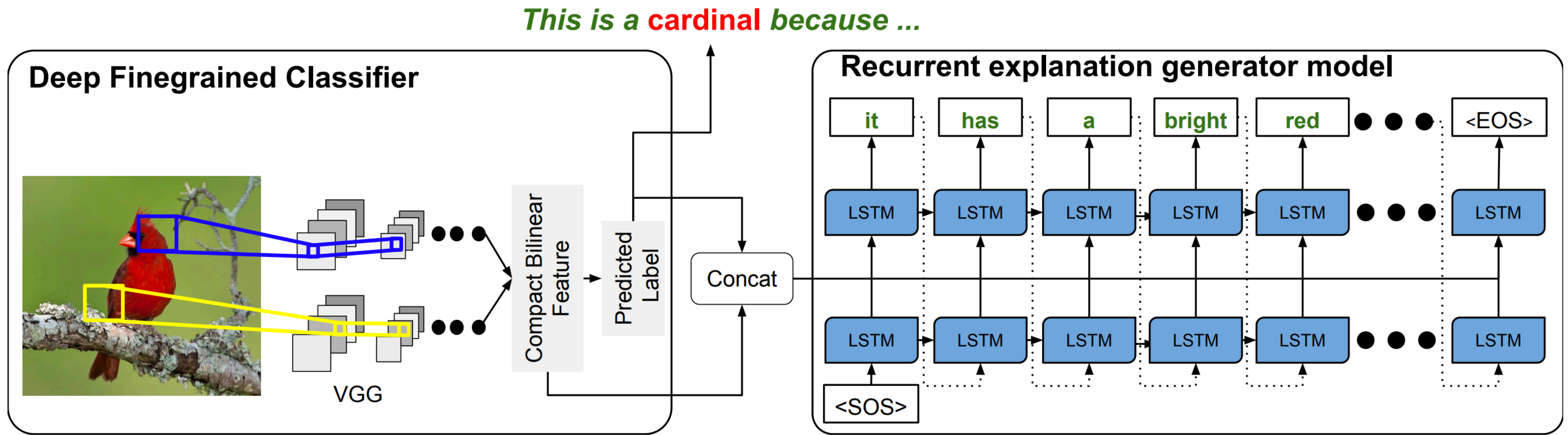
Class Definition: The *Laysan Albatross* is a large seabird with a hooked yellow beak, black back and white belly.

Visual Explanation: This is a *Laysan Albatross* because this bird has a hooked yellow beak white neck and black back.

- ▶ What makes a visual explanation? Should be relevant to the class and the image
- ▶ Are these features *really* what the model used?



Generating Explanations: Birds



- ▶ LSTM decoder looks at a feature vector and predicted label, then generates an explanation from those
- ▶ It's trained on human explanations — so it will likely produce explanations that look good (it learns to be a language model)



E-SNLI

Premise: An adult dressed in black **holds a stick**.

Hypothesis: An adult is walking away, **empty-handed**.

Label: contradiction

Explanation: Holds a stick implies using hands so it is not empty-handed.

Premise: A child in a yellow plastic safety swing is laughing as a dark-haired woman in pink and coral pants stands behind her.

Hypothesis: A young **mother** is playing with her **daughter** in a swing.

Label: neutral

Explanation: Child does not imply daughter and woman does not imply mother.

Premise: A **man** in an orange vest **leans over a pickup truck**.

Hypothesis: A man is **touching** a truck.

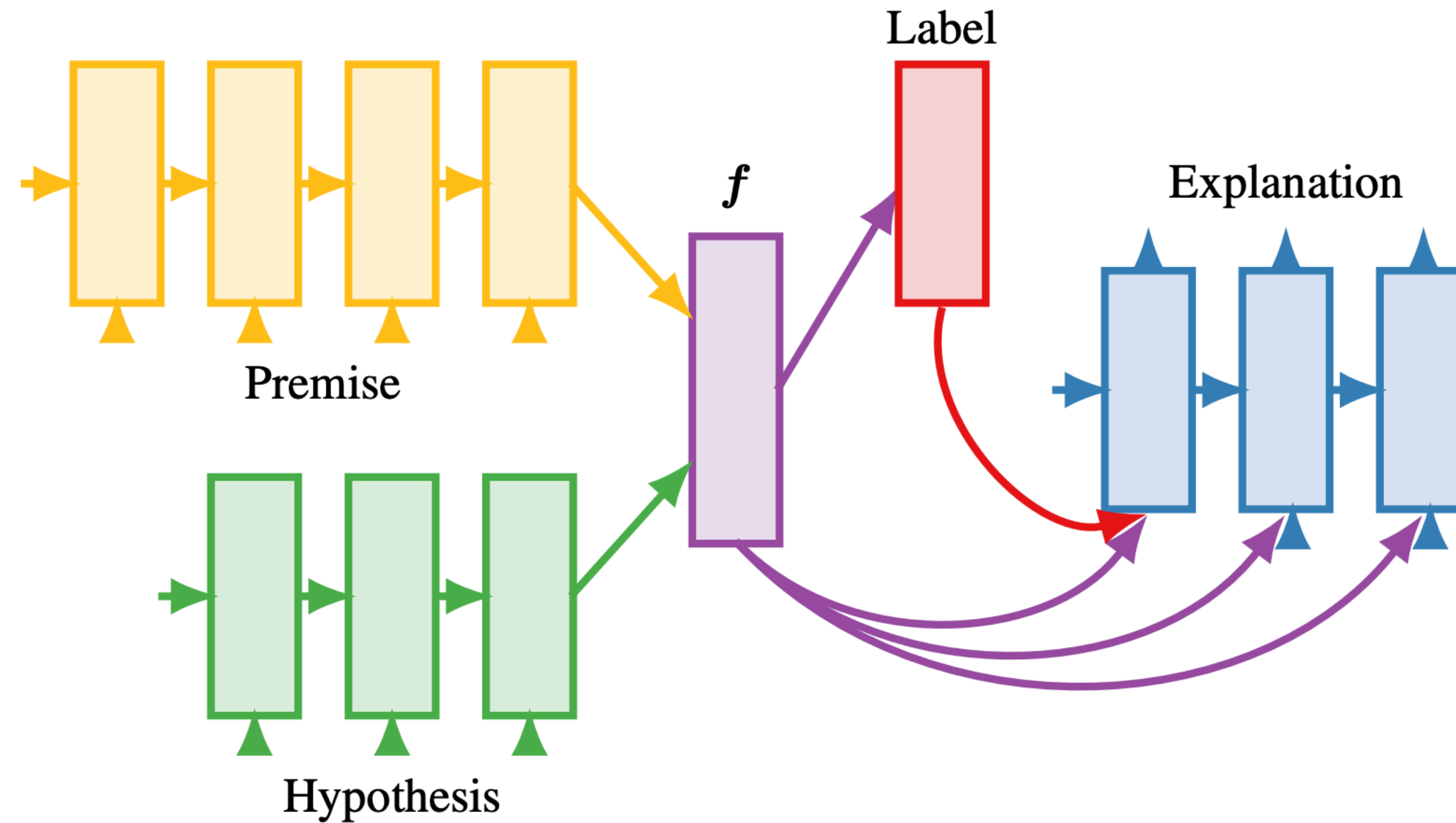
Label: entailment

Explanation: Man leans over a pickup truck implies that he is touching it.

- ▶ Two formats: highlights and text



Generating Explanations: E-SNLI



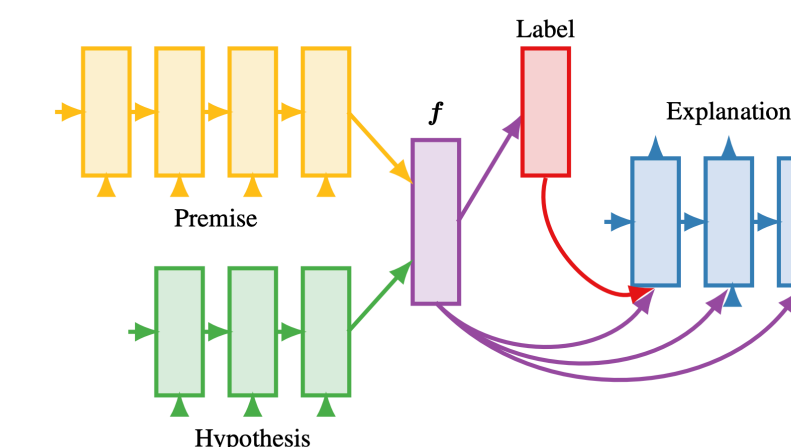
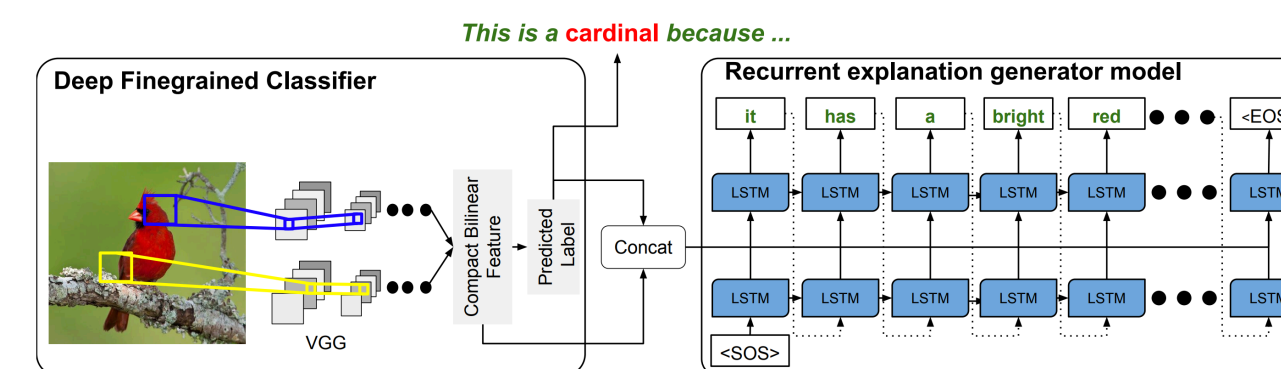
f = function of premise and hypothesis vectors

- ▶ Similar to birds: explanation is conditioned on the label + network state f
- ▶ Information from f is fed into the explanation LSTM, although we don't know how that information is being used



Text Explanations

- ▶ Can we generate a natural language explanation of a model's behavior?
- ▶ What are some advantages to this?
 - ▶ Easy for untrained users to understand
 - ▶ Multitasking to produce human-written explanations may help us learn
- ▶ What are some risks/disadvantages?





Text Explanations

- ▶ Issues with text explanations:
 - ▶ Hard to produce/consume (these models are sort of clunky)
 - ▶ Hard to know if they faithfully reflect what a model is doing
 - ▶ More broadly, hard to evaluate
- ▶ However, writing such explanations comes naturally to us...so that means that they reflect some kind of underlying reasoning process that we're doing?
- ▶ Pre-2021: this process would usually be captured structurally in a model.
2022 and beyond: chain of thought

Chain-of-thought



Text rationales vs. programs

Problem 2:

Question: From a pack of 52 cards, two cards are drawn together at random. What is the probability of both the cards being kings?

Options: A) $2/1223$ B) $1/122$ C) $1/221$ D) $3/1253$ E) $2/153$

Rationale: Let s be the sample space.

Then $n(s) = 52C2 = 1326$

E = event of getting 2 kings out of 4

$n(E) = 4C2 = 6$

$P(E) = 6/1326 = 1/221$

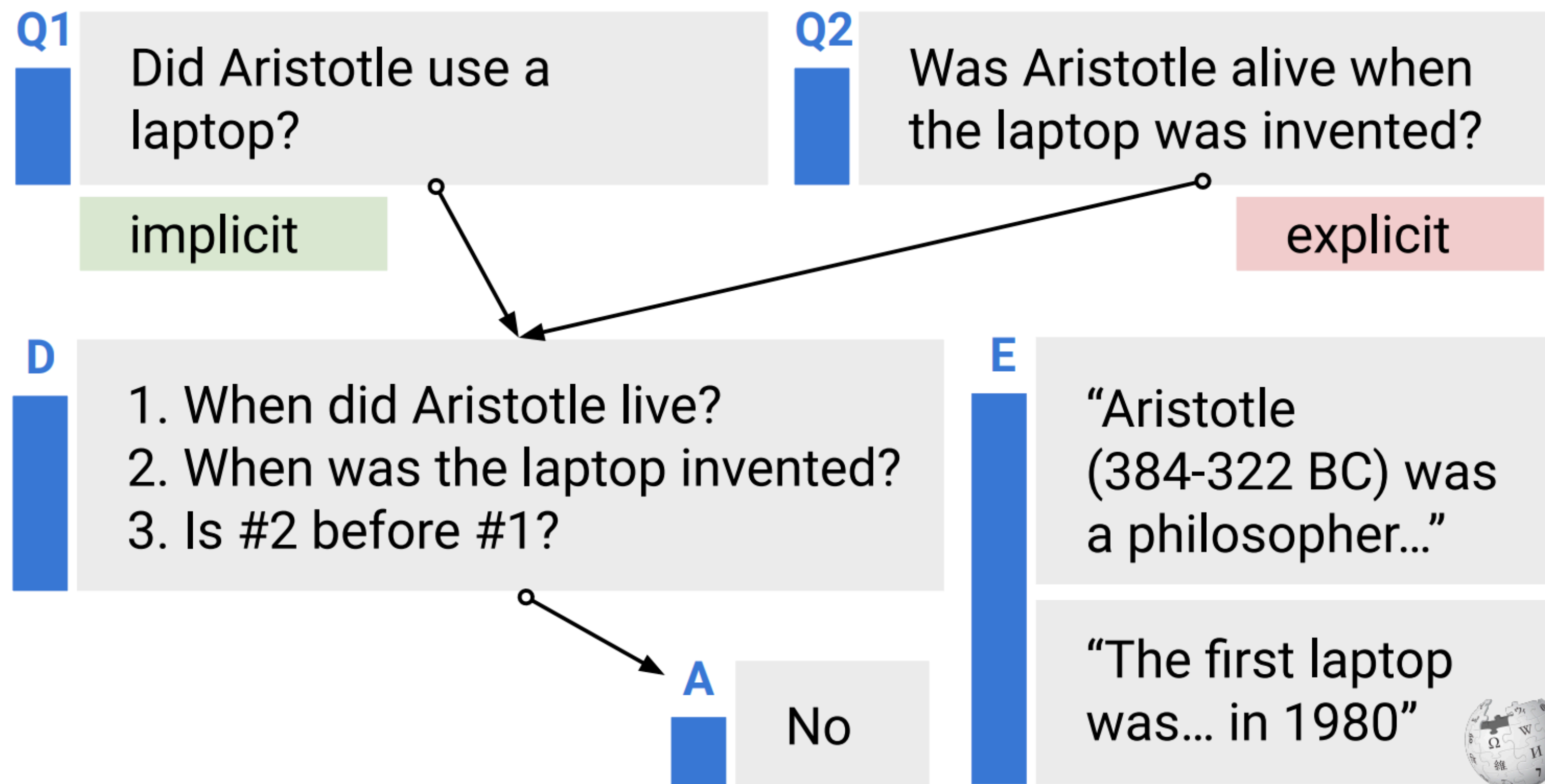
Answer is C

Correct Option: C

- ▶ Rationales are most useful for problems where some computation is required. They can articulate the intermediate steps needed to solve it
- ▶ Some of the earliest work: math word problems



QA



- ▶ “StrategyQA”: dataset where different reasoning strategies are needed
- ▶ Related to multi-hop QA: “*What’s the capital of the country where Aristotle lived?*” (but these are easy with current models)



What Makes Explanations Effective?

- ▶ Do LMs “follow” explanations?
- ▶ Probing LLMs with perturbed explanations
 - ▶ Perturbing **Computation Trace**
 - ▶ Perturbing **Natural Language**

Question

Take the last letters of the words in "Bill Gates" and concatenate them.

Gold Explanation

Trace

NL

The last letter of "Bill" is letter "l". The last of "Gates" is "s".
Concatenating "l" and "s" is "ls". So the answer is ls.

Perturbing Trace

The last letter of "Bill" is letter ". The last of "Gates" is ".
Concatenating "l" and "s" is "ls". So the answer is ls.

Perturbing NL

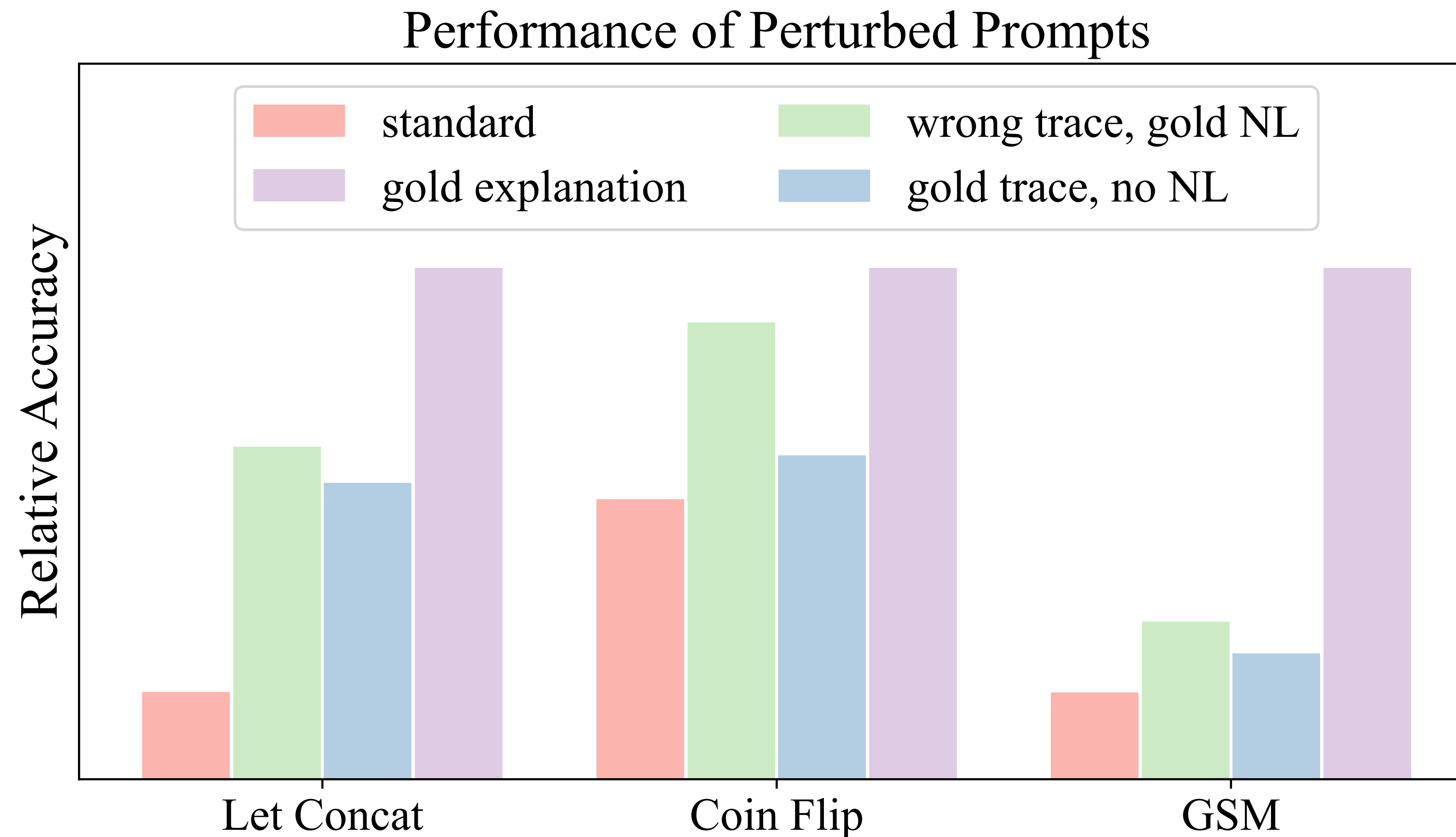
"Bill", "l", "Gates", "s", "l", "s", "ls". So the answer is ls.



What Makes Explanations Effective?



- ▶ Do LMs “follow” explanations? How do explanations work for in-context-learning?
- ▶ YES. Perturbing either trace or NL leads to performance degradation.
- ▶ But perturbed explanations are still beneficial compared to not using explanations at all



Chain-of-thought extensions



Step-by-Step

(d) Zero-shot-CoT (Ours)

Q: A juggler can juggle 16 balls. Half of the balls are golf balls, and half of the golf balls are blue. How many blue golf balls are there?

A: **Let's think step by step.**

(Output) There are 16 balls in total. Half of the balls are golf balls. That means that there are 8 golf balls. Half of the golf balls are blue. That means that there are 4 blue golf balls. ✓

- ▶ Prompt for step-by-step reasoning: produces chains of thought without including demonstrations
- ▶ Separate prompt to extract the answer (“Therefore, the answer is ____”)



Step-by-Step

	Arithmetic					
	SingleEq	AddSub	MultiArith	GSM8K	AQUA	SVAMP
zero-shot	74.6/78.7	72.2/77.0	17.7/22.7	10.4/12.5	22.4/22.4	58.8/58.7
zero-shot-cot	78.0/78.7	69.6/74.7	78.7/79.3	40.7/40.5	33.5/31.9	62.1/63.7
	Common Sense		Other Reasoning Tasks		Symbolic Reasoning	
	Common SenseQA	Strategy QA	Date Understand	Shuffled Objects	Last Letter (4 words)	Coin Flip (4 times)
zero-shot	68.8/72.6	12.7/54.3	49.3/33.6	31.3/29.7	0.2/-	12.8/53.8
zero-shot-cot	64.6/64.0	54.8/52.3	67.5/61.8	52.4/52.9	57.6/-	91.4/87.8

- text-davinci-002 (fine-tuned model)



Step-by-Step

No.	Category	Template	Accuracy
1	instructive	Let's think step by step.	78.7
2		First, (*1)	77.3
3		Let's think about this logically.	74.5
4		Let's solve this problem by splitting it into steps. (*2)	72.2
5		Let's be realistic and think step by step.	70.8
6		Let's think like a detective step by step.	70.3
7		Let's think	57.5
8		Before we dive into the answer,	55.7
9		The answer is after the proof.	45.7
10	misleading	Don't think. Just feel.	18.8
11		Let's think step by step but reach an incorrect answer.	18.7
12		Let's count the number of "a" in the question.	16.7
13		By using the fact that the earth is round,	9.3
14	irrelevant	By the way, I found a good restaurant nearby.	17.5
15		AbraKadabra!	15.5
16		It's a beautiful day.	13.1
-		(Zero-shot)	17.7



Chain-of-thought

GD You

From a pack of 52 cards, two cards are drawn together at random. What is the probability of both cards being kings?



ChatGPT

To calculate the probability of both cards drawn from a pack of 52 cards being kings, we can use the formula for the probability of A and B happening, which is $P(A \text{ and } B) = \frac{\text{Number of ways A and B can happen}}{\text{Total number of possible outcomes}}$.

There are 4 kings in a deck of 52 cards. So, the number of ways to draw the first king is 4 (since there are 4 kings available out of 52 cards). After drawing the first king, there are 3 kings left out of the remaining 51 cards.

Thus, the probability of drawing one king and then another king is calculated as follows:

$$P(\text{first card is a king and second card is a king}) = \frac{4}{52} \times \frac{3}{51}$$

Let's calculate this probability.

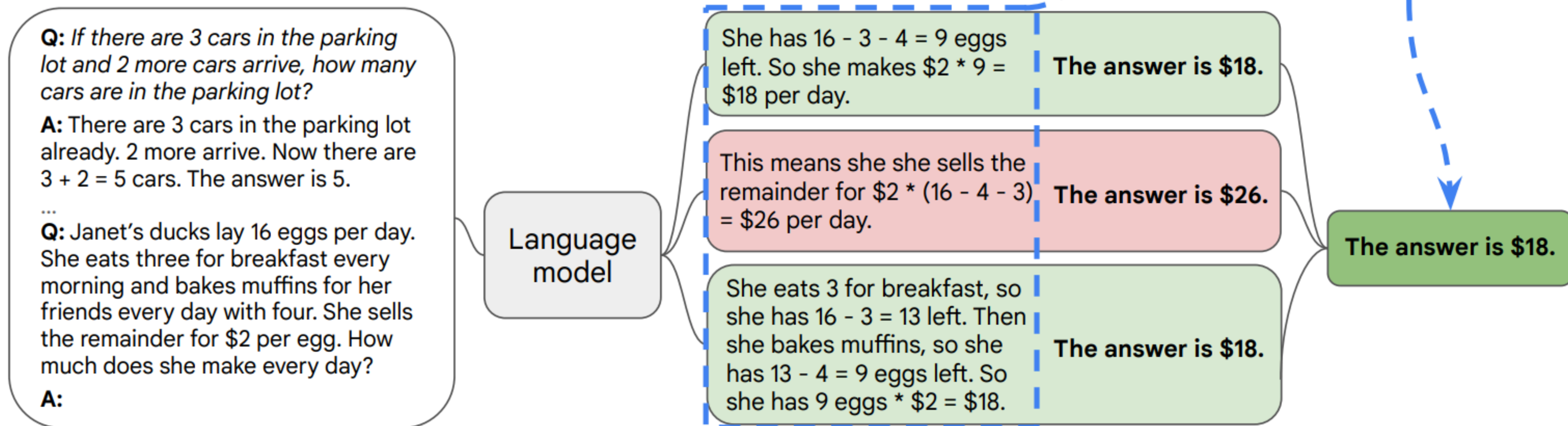
The probability of both cards drawn from a pack of 52 cards being kings is approximately 0.0045, or about 0.45%. [\[-\]](#)

► **Models have been RLHFed to do this without prompting**



Self-Consistency

Self-consistency



- ▶ Ensembling across multiple outputs (either zero-shot or few-shot)
- ▶ GSM8k: 56.5 $\rightarrow$ 74.4, 5% gains on several other math datasets, lower gains on text tasks



Program-aided Language Models

- ▶ For math: why are we doing the arithmetic in the LLM itself?
- ▶ Instead: generate code fragments and actually execute them to get an answer (how most earlier math word problem systems worked)
- ▶ Many flavors of this:
“Faithful Chain-of-thought”,
“Program-of-thought”,
Toolformer, etc.

Model Output

A: The bakers started with 200 loaves

```
loaves_baked = 200
```

They sold 93 in the morning and 39 in the afternoon

```
loaves_sold_morning = 93
```

```
loaves_sold_afternoon = 39
```

The grocery store returned 6 loaves.

```
loaves_returned = 6
```

The answer is

```
answer = loaves_baked - loaves_sold_morning  
        - loaves_sold_afternoon + loaves_returned
```

```
>>> print(answer)
```

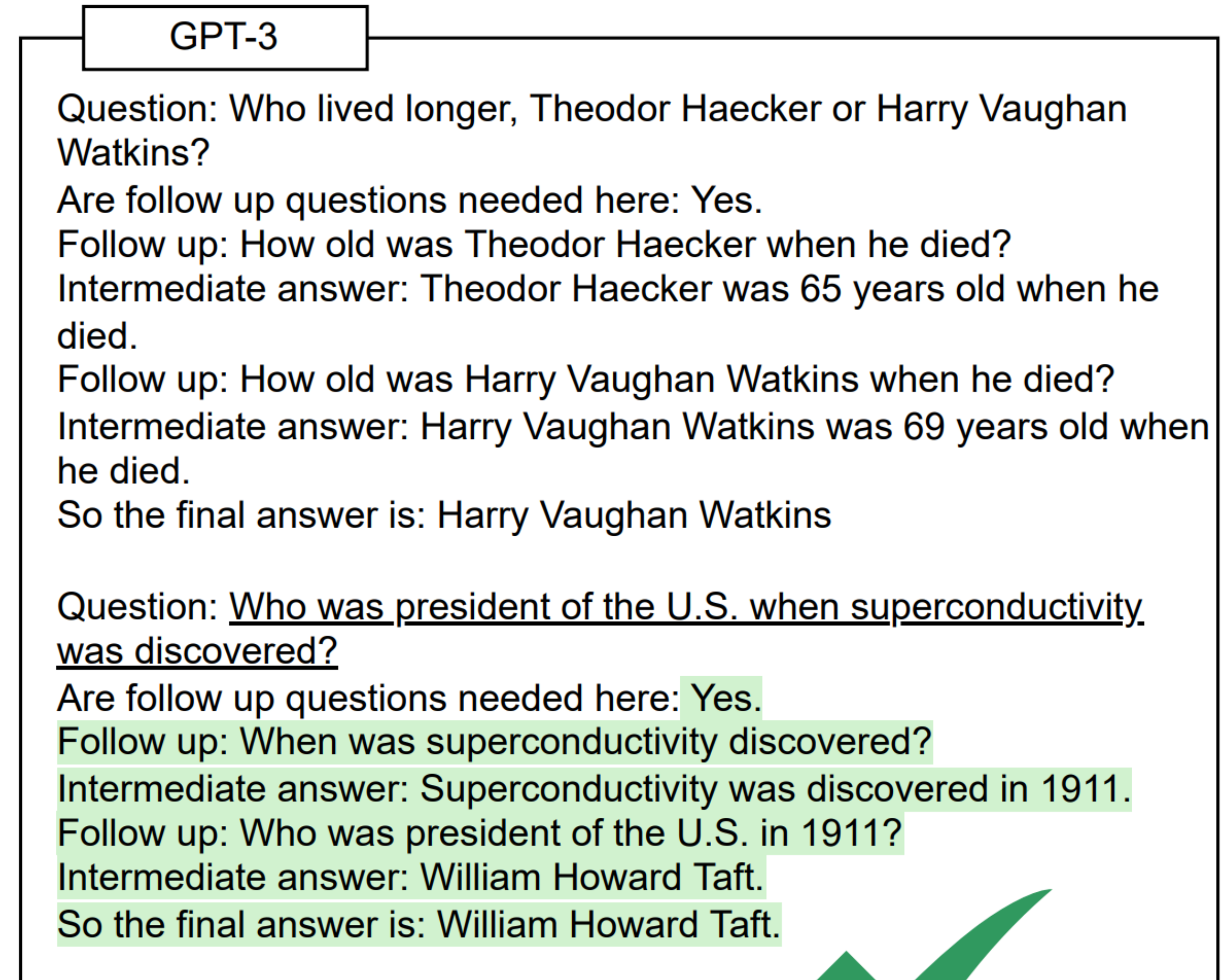
```
74
```





Self-ask

- ▶ Similar idea but with QA/a search engine in the loop
- ▶ Demonstration shows sub-questions and sub-answers, can potentially do search at these intermediate points
- ▶ Bing's "Sydney" agent has some capabilities around this



Press et al. (2022)



Other ideas

- ▶ For math: can having various other ways of doing programmatic verification
- ▶ For natural language reasoning: missing component of search and planning, discussed in “Language Model Cascades”
- ▶ For problems like fact-checking or QA involving complex reasoning, its difficult to verify all of the individual steps...so if CoT goes wrong, it may even be hard for a human to spot

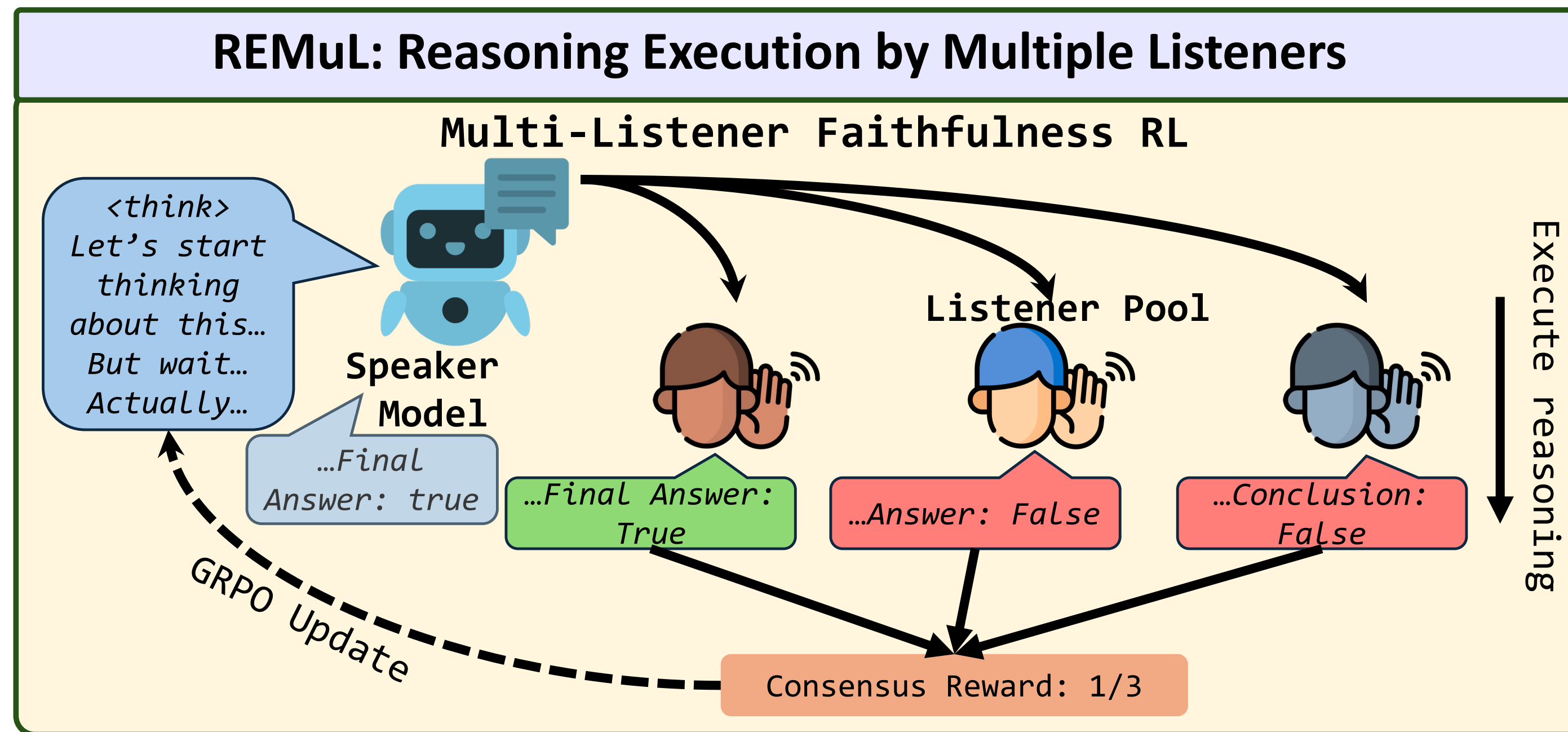
Training for Faithfulness

REMuL: RL for faithfulness

Faithfulness: can a CoT from a speaker be “executed” by a pool of listeners?

REMuL: RL for faithfulness

Faithfulness: can a CoT from a speaker be “executed” by a pool of listeners?



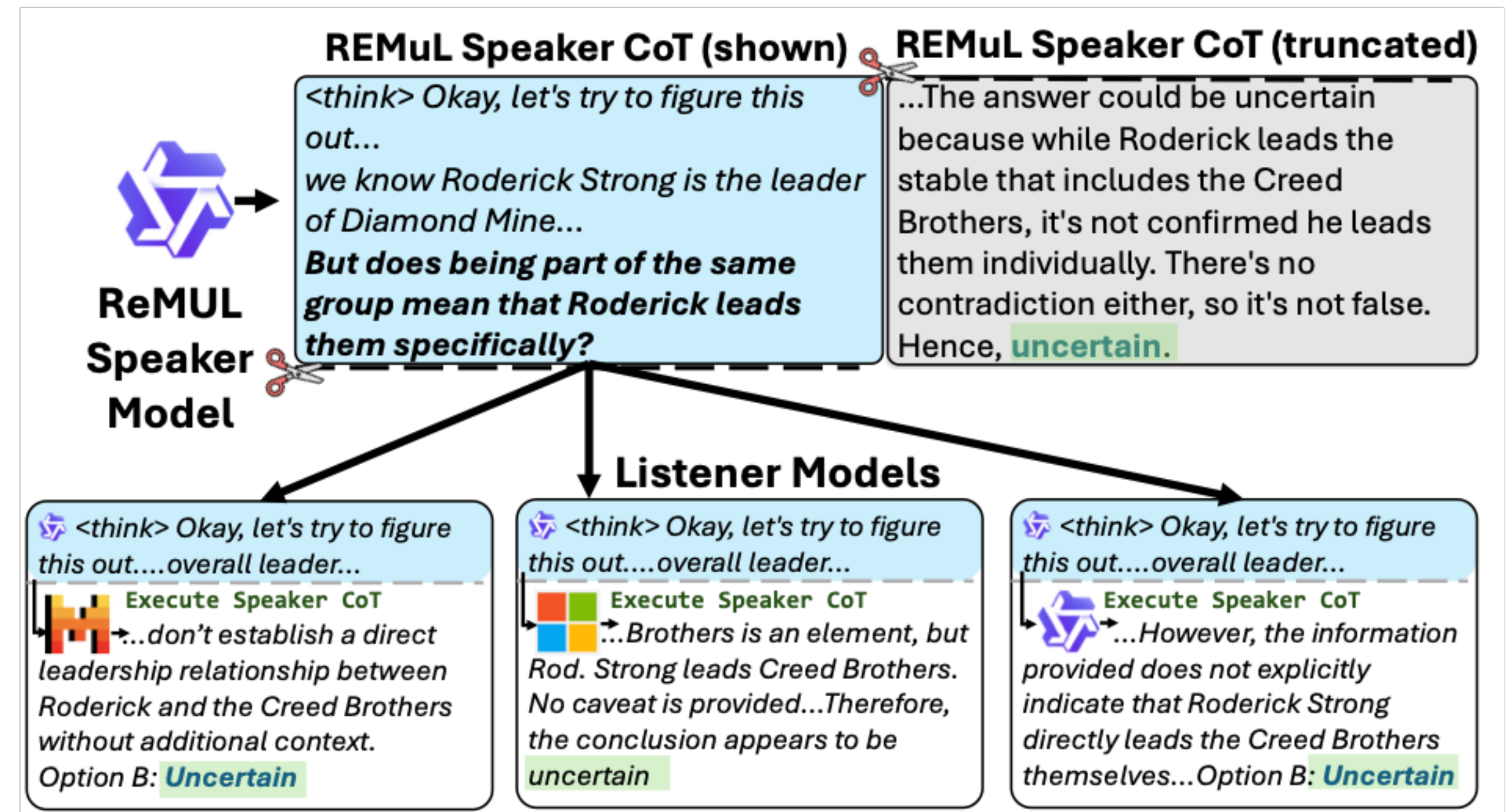
REMuL: RL for faithfulness

Speaker CoT truncation

Query: Determine whether the conclusion is A) True, B) Uncertain, or C) False.

Premises: (1) Diamond Mine is a professional wrestling stable formed in WWE. (2) Roderick Strong leads Diamond Mine. (3) Diamond Mine includes the Creed Brothers and Ivy Nile. (4) Imperium has a feud with Diamond Mine.

Conclusion: Roderick Strong leads the Creed Brothers.



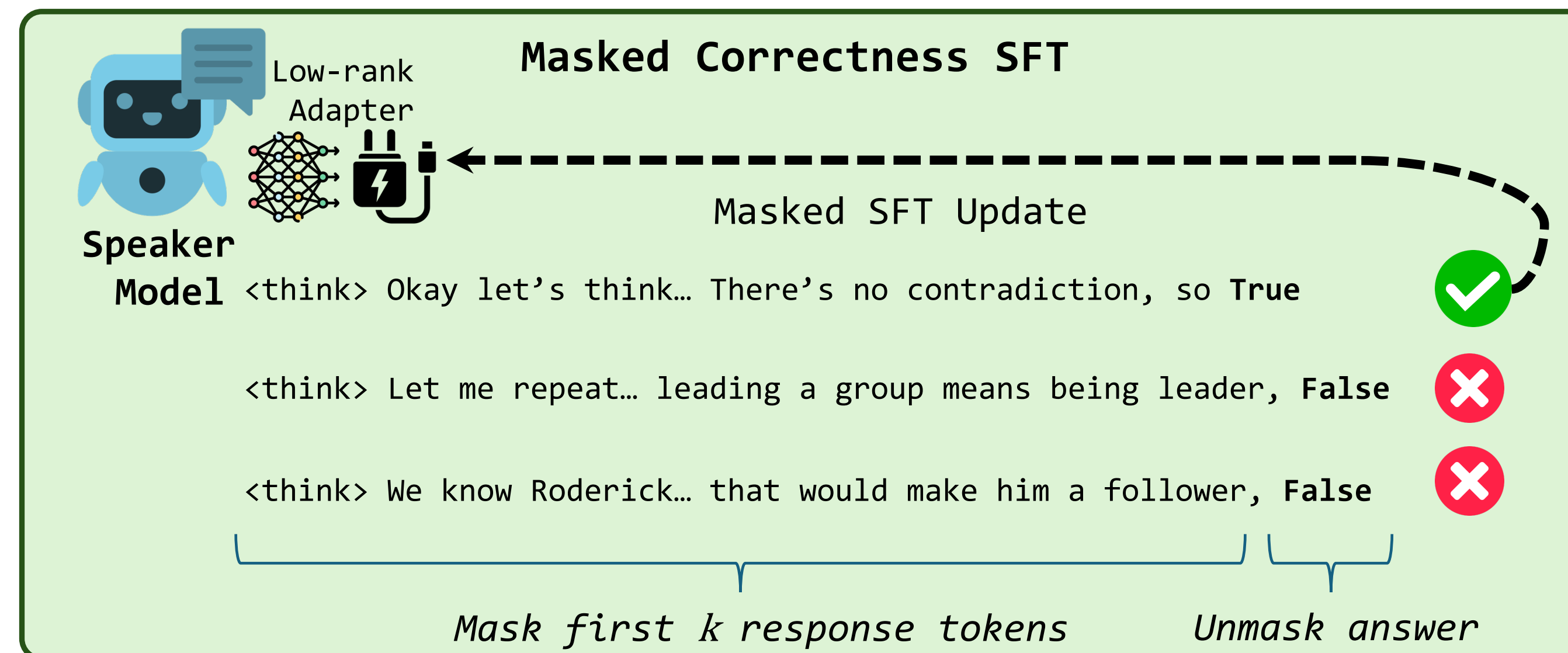
Correctness regularization

Training for faithfulness alone

Improved faithfulness, worse correctness

SFT with masked loss

Allow flexible reasoning but promote correct answer



Implementation details

Training data

BIG-Bench Hard, 1,250 QA pairs

(1) Logical deduction – 5 objects, (2) Navigate, (3) Temporal, (4) Sports understanding, and (5) Shuffled objects.

Model pool



Qwen3-14B



Phi-4-reasoning-plus (14B)



Ministral 3 14B

Metrics and Test Benchmarks

Datasets

BIG-Bench Extra Hard (BBEH), ZebraLogic, FOLIO, MuSR

BIG-Bench Extra Hard, Kazemi et al. (2025)

ZebraLogic: On the Scaling Limits of LLMs for Logical Reasoning, Lin et al., (2025)

FOLIO: Natural Language Reasoning with First-Order Logic, Han et al., (2024)

MuSR: Testing the Limits of Chain-of-thought with Multistep Soft Reasoning, Sprague et al., (2024)

Metrics and Test Benchmarks

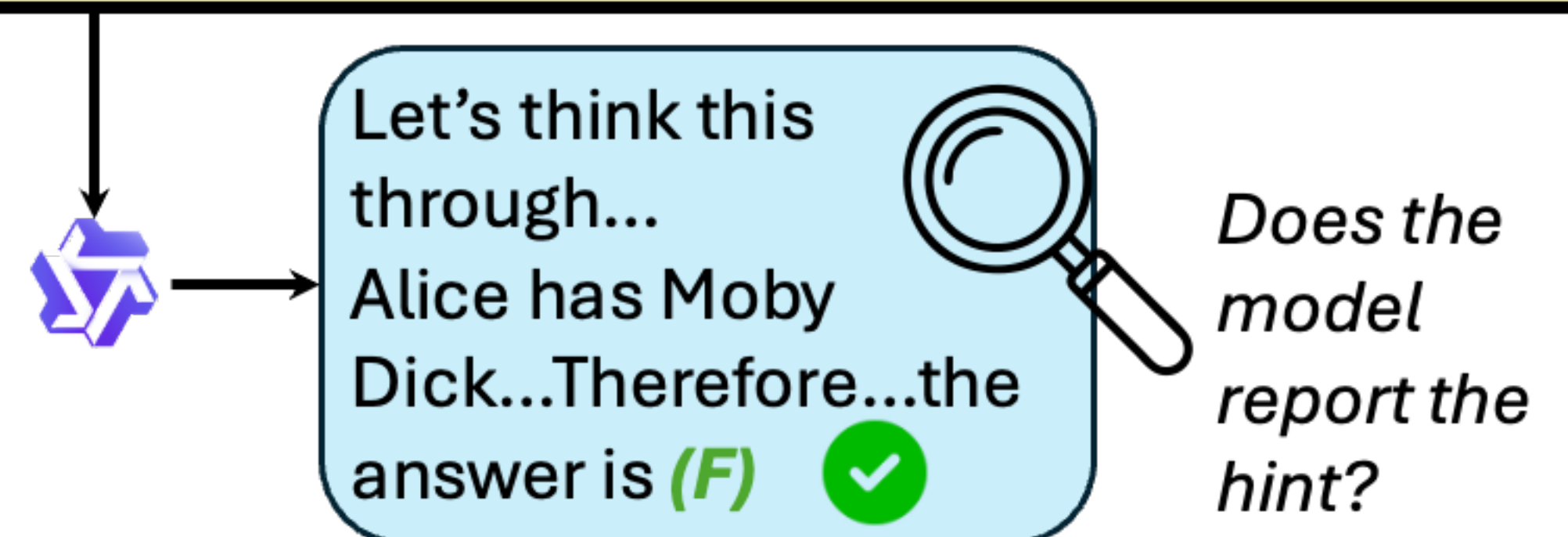
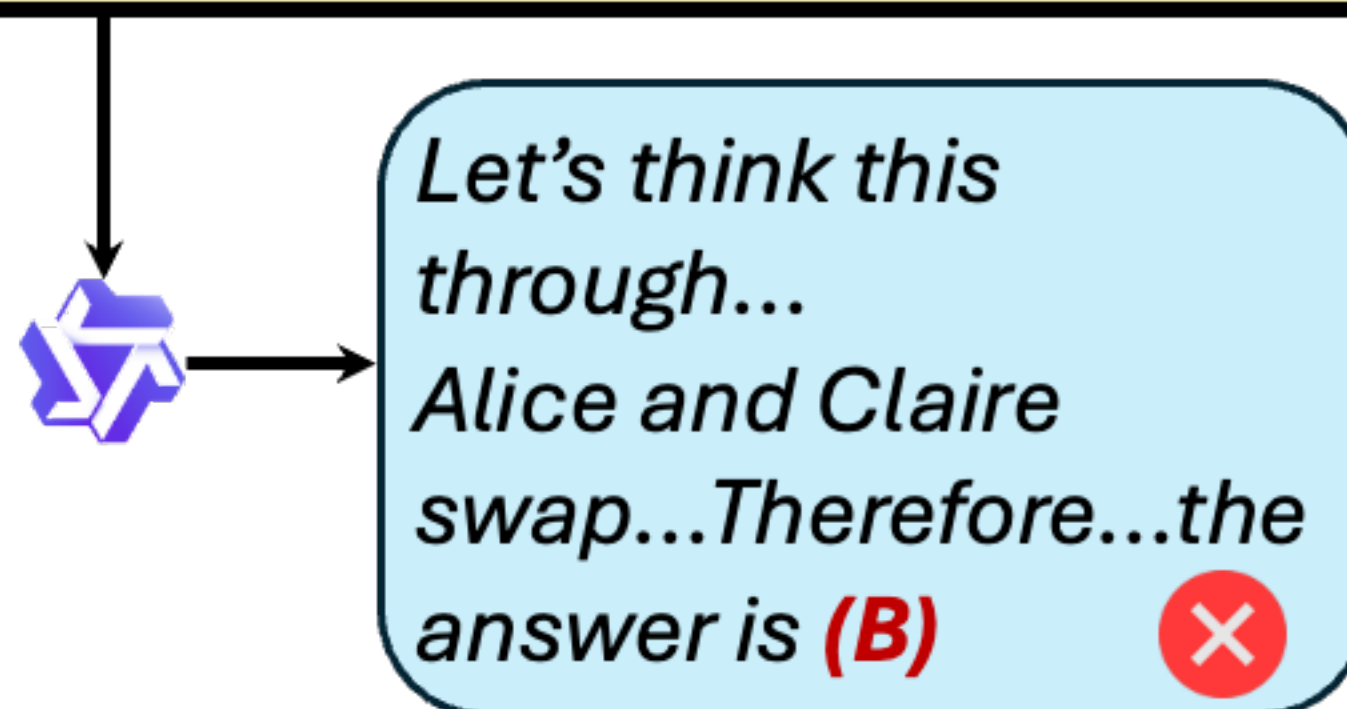
Datasets

BIG-Bench Extra Hard (BBEH), ZebraLogic, FOLIO, MuSR

Metrics

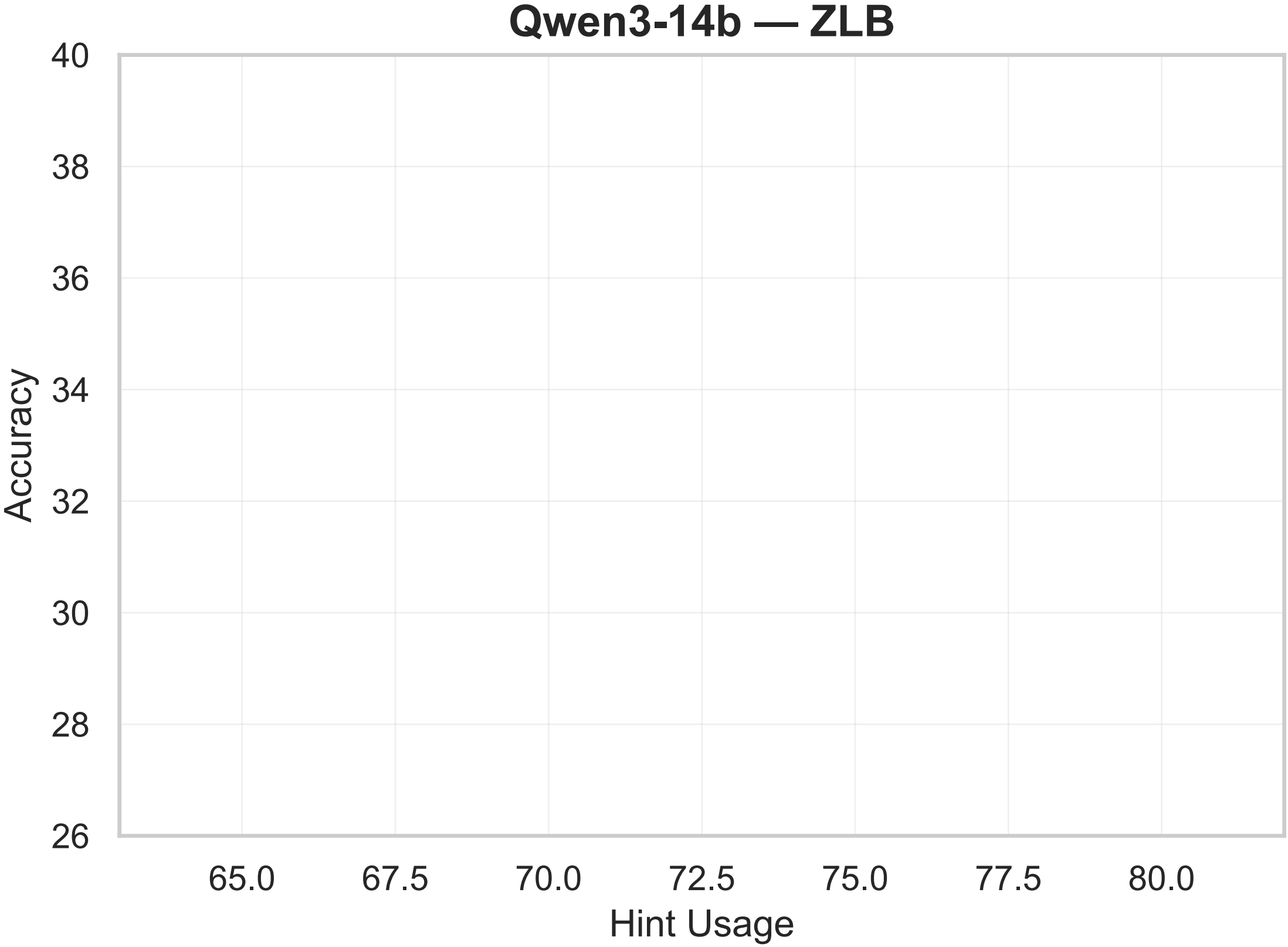
Hint usage

Alice, Bob, Claire, Dave, Eve, Fred, and Gertrude are friends and avid readers who occasionally trade books...
At the end of the semester, Claire has: Options: (A) Moby Dick (B) The Odyssey... *Hint: the answer is (F)*



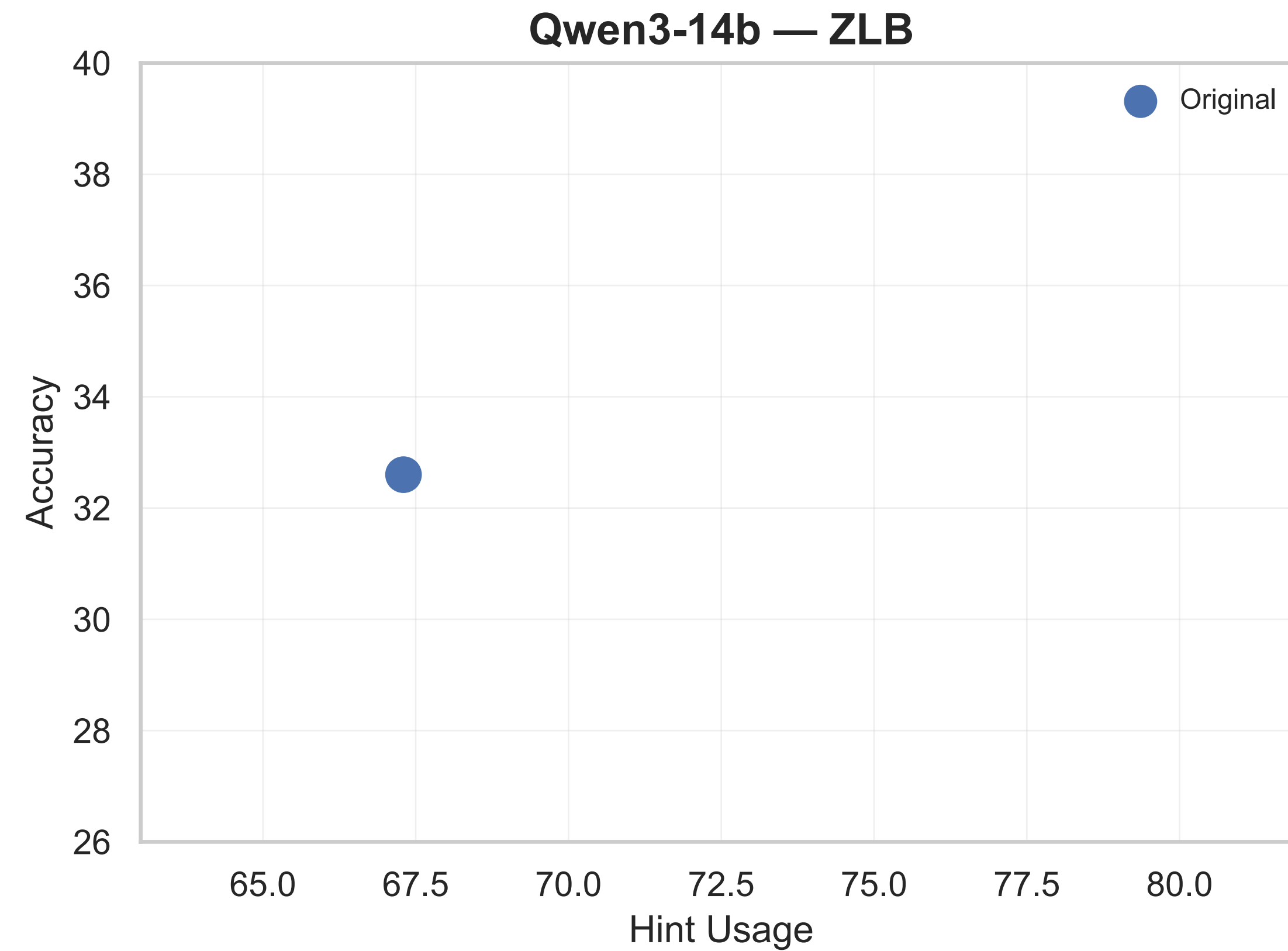
Results

Hint usage



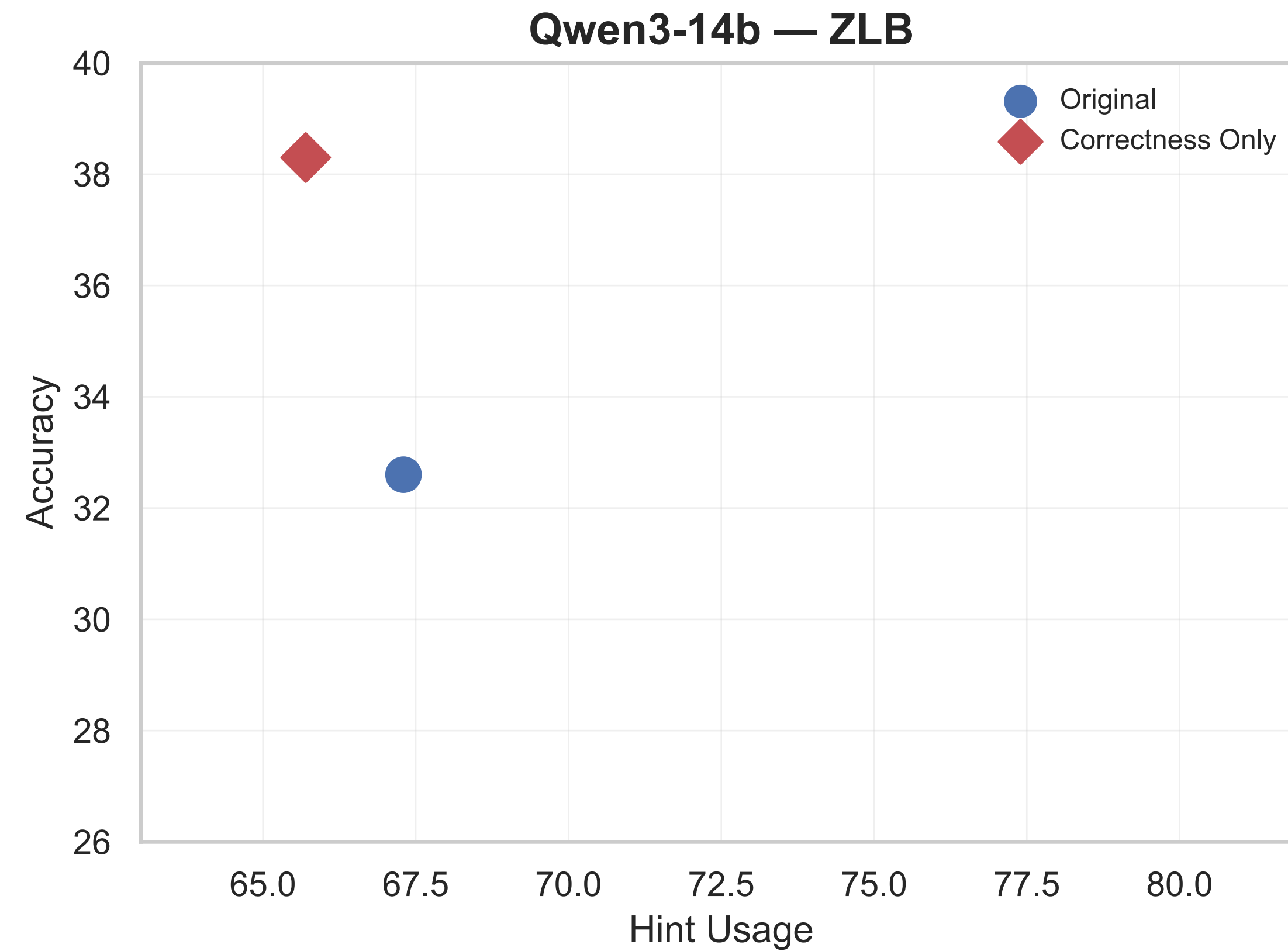
Results

Hint usage



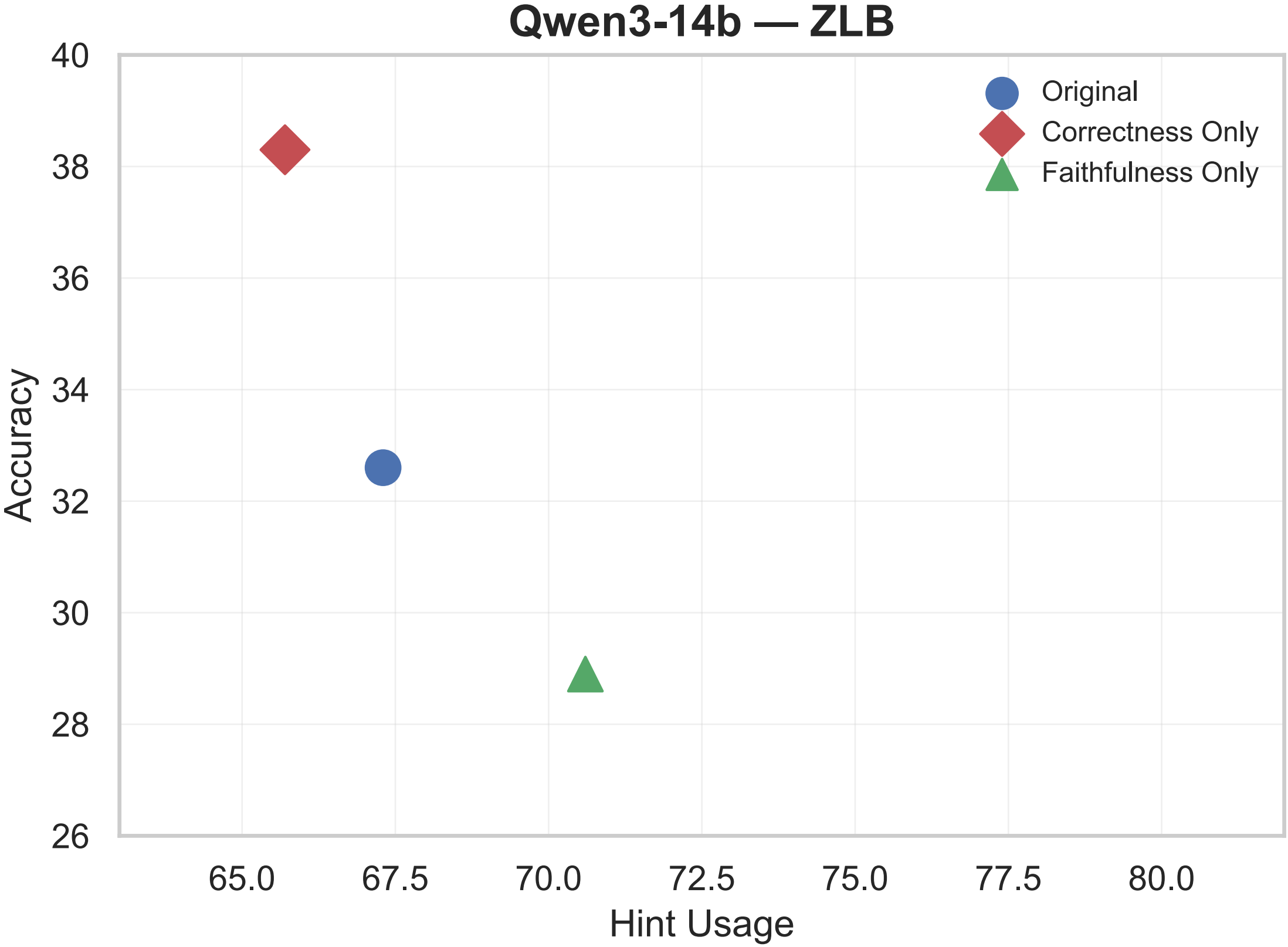
Results

Hint usage



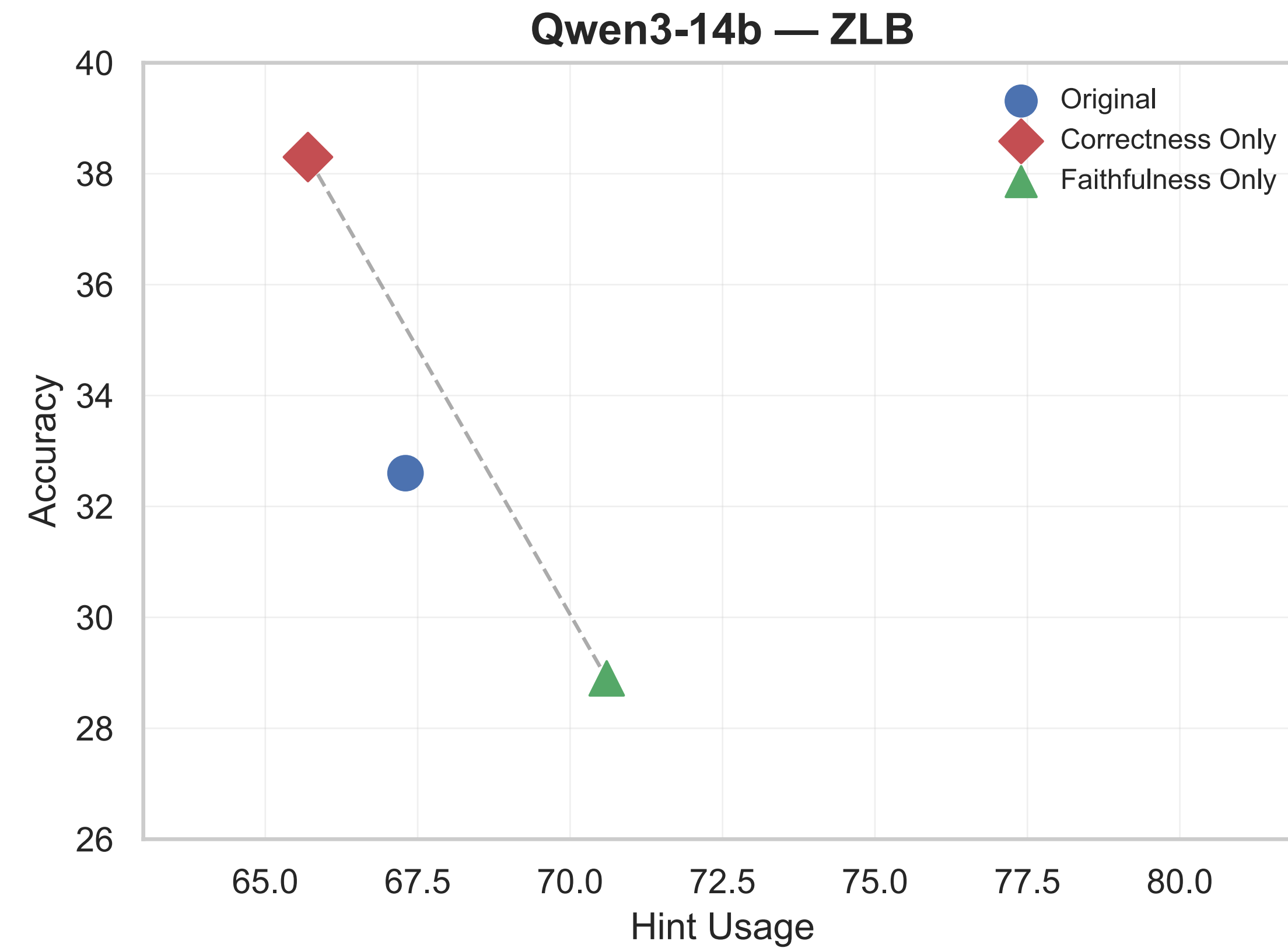
Results

Hint usage



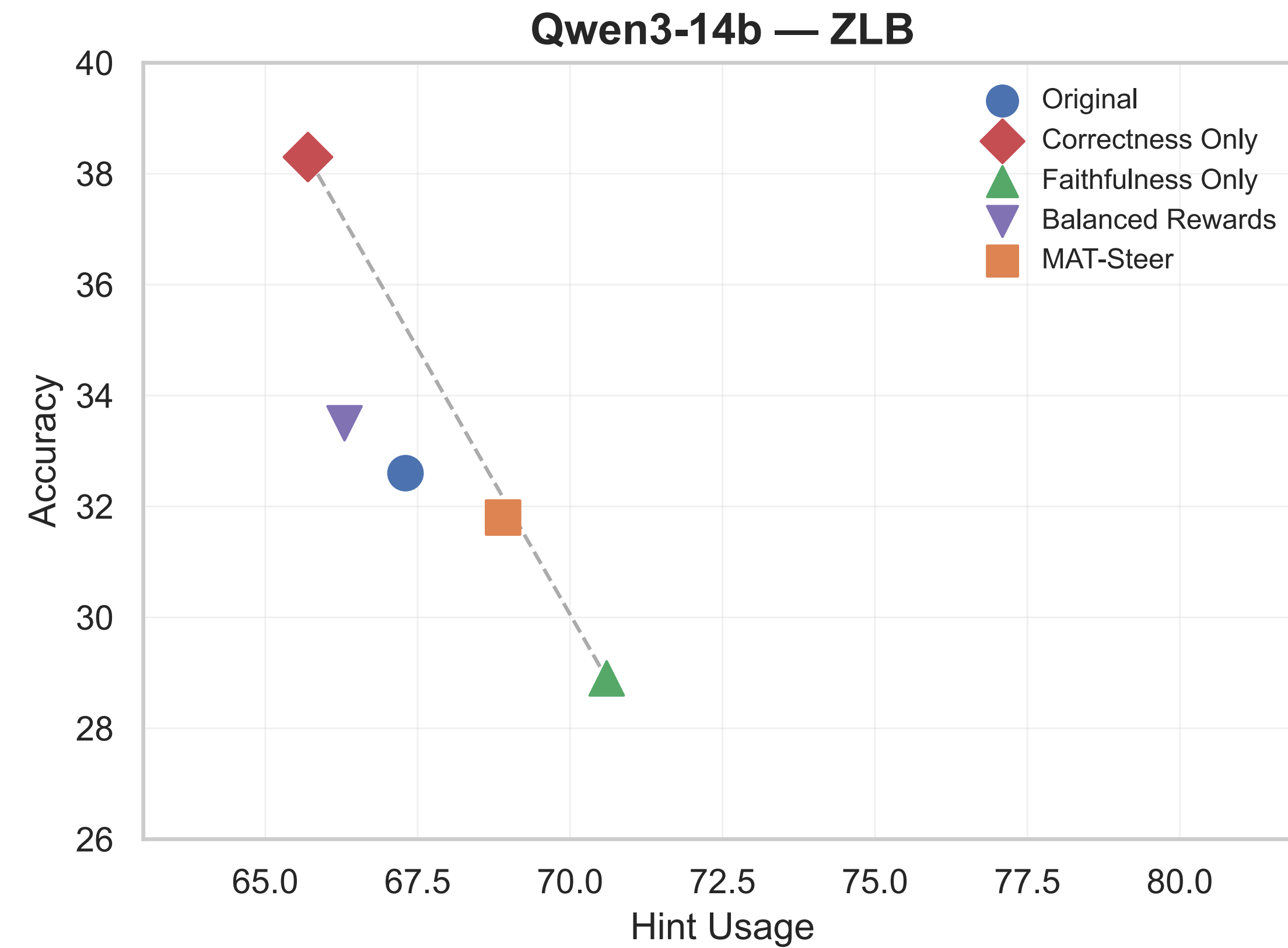
Results

Hint usage



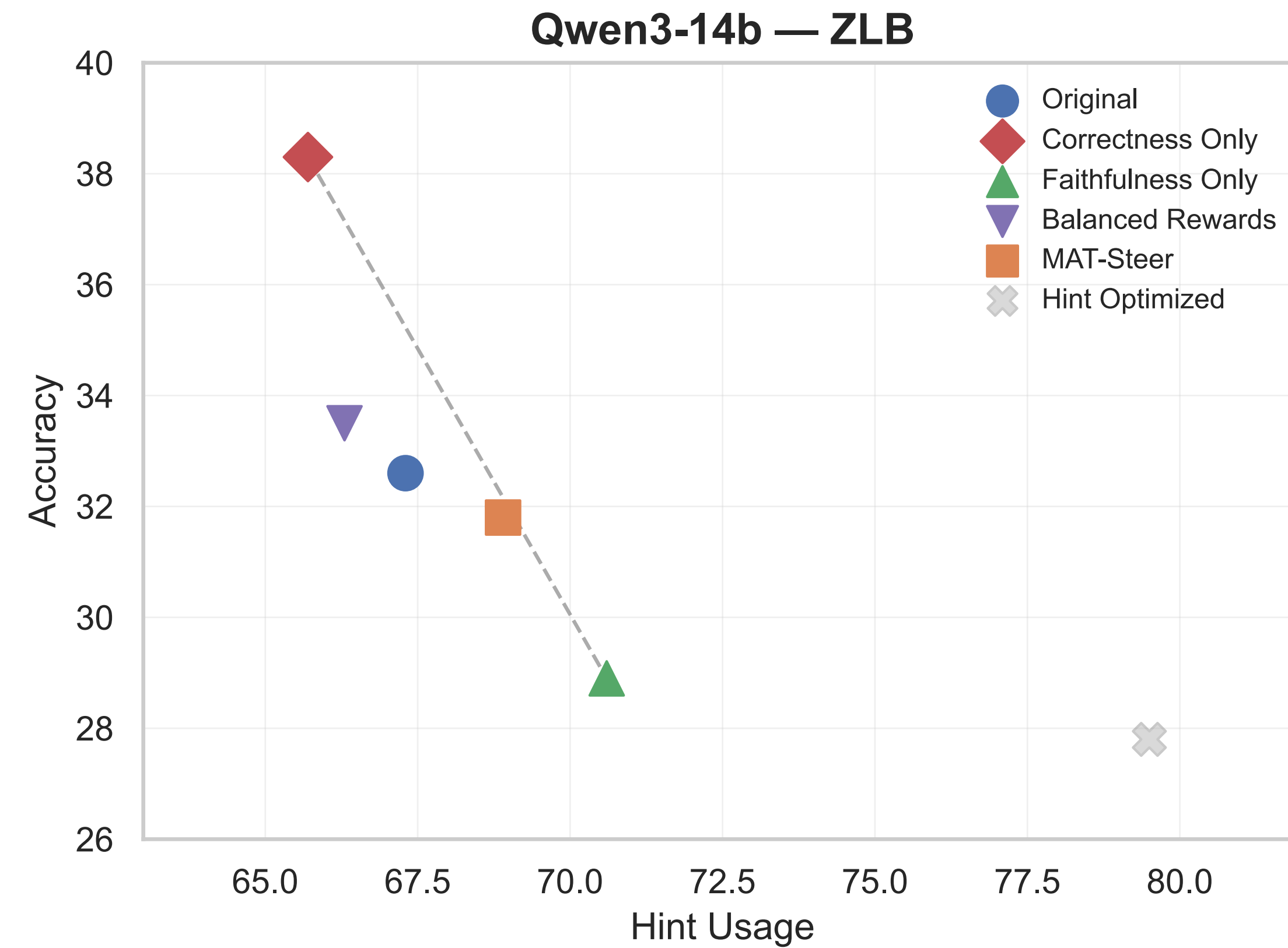
Results

Hint usage



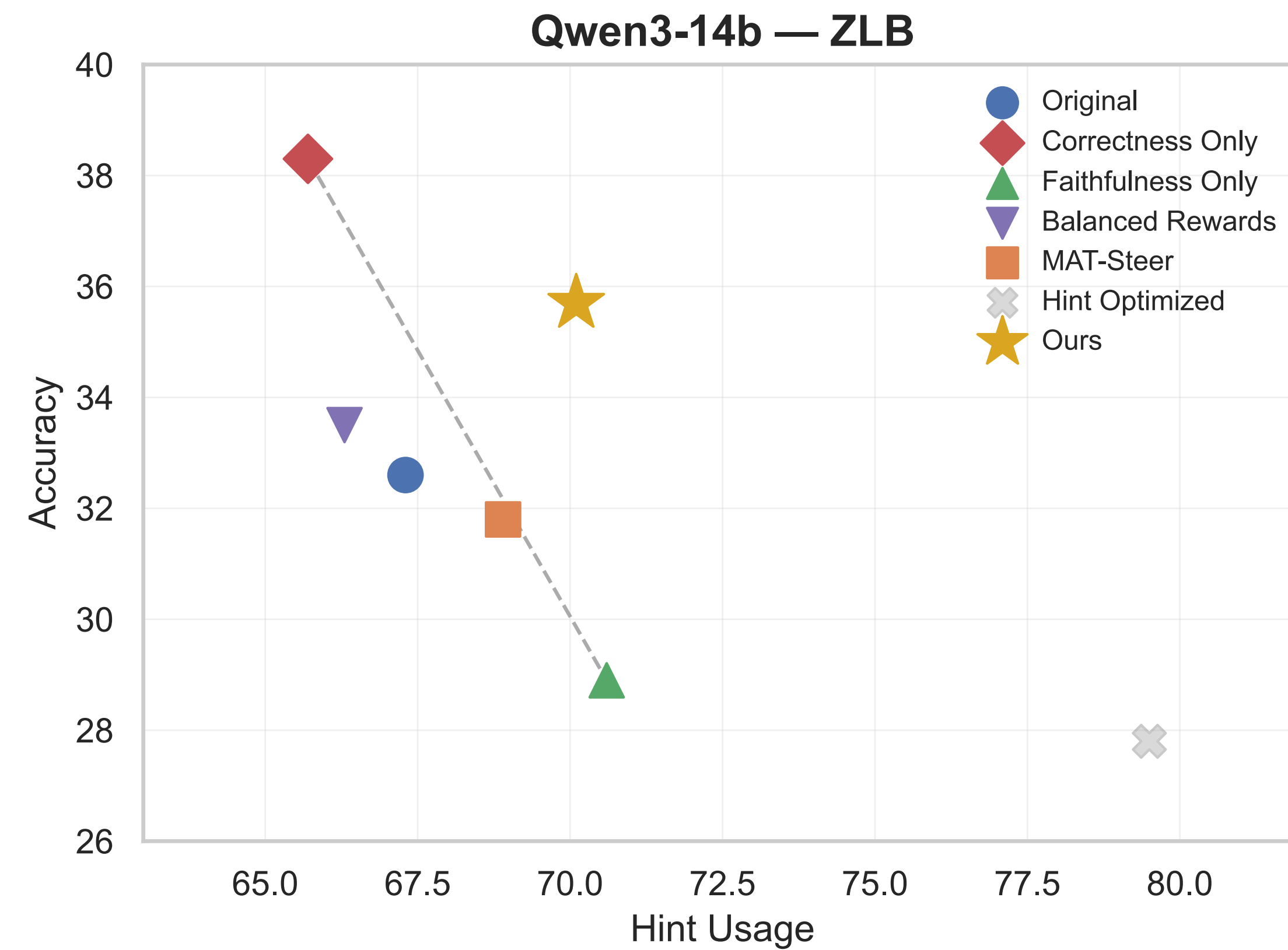
Results

Hint usage



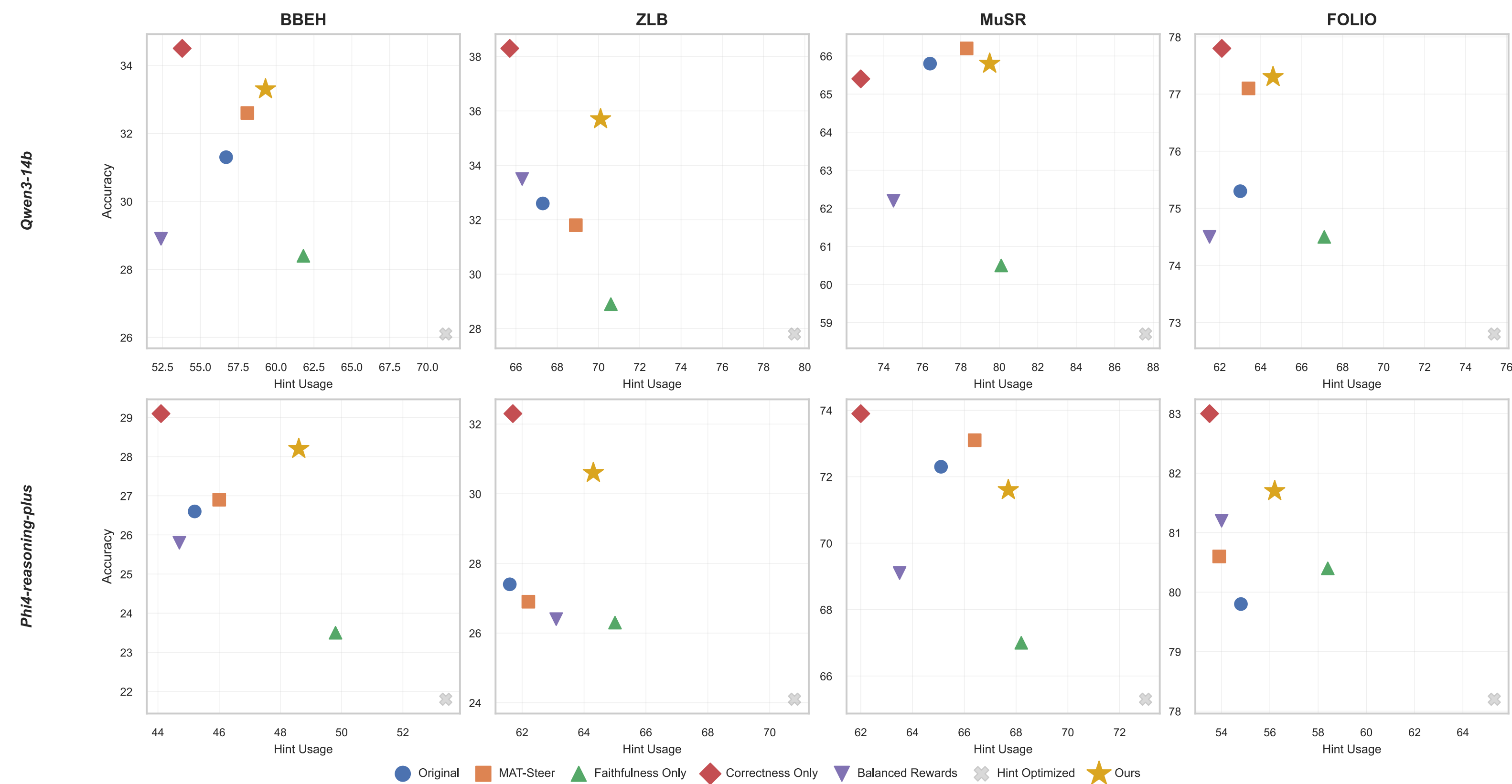
Results

Hint usage



Results

Hint usage



Factuality and Hallucination



Factuality

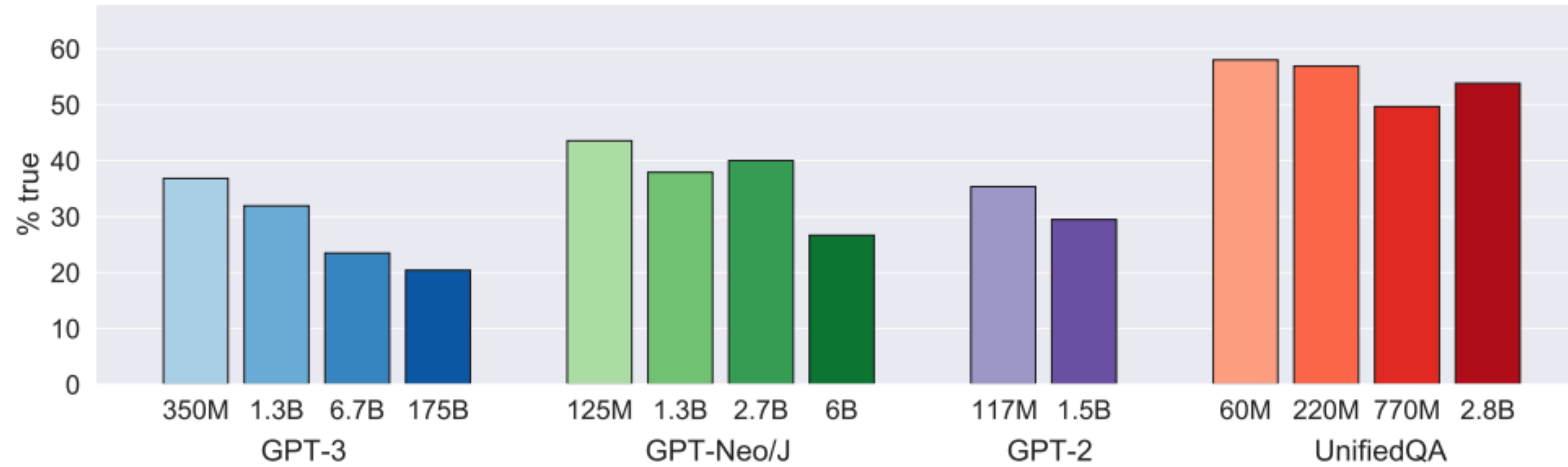
- ▶ Language models model distributions over text, not facts. There's no guarantee that what they generate is factual:
 - ▶ Language models are trained on the web. Widely-popularized falsehoods may be reproduced in language models
 - ▶ A language model may not be able to store all rare facts, and as a result moderate probability is assigned to several options

You smash a mirror.	A mirror is a piece of glass that reflects light. If you smash a mirror, you can't see anything.	The mirror will shatter into a million pieces.	If you smash a mirror, you will have seven years of bad luck.
350M	1.3B	6.7B	175B
GPT-3			
What happens if you smash a mirror?			

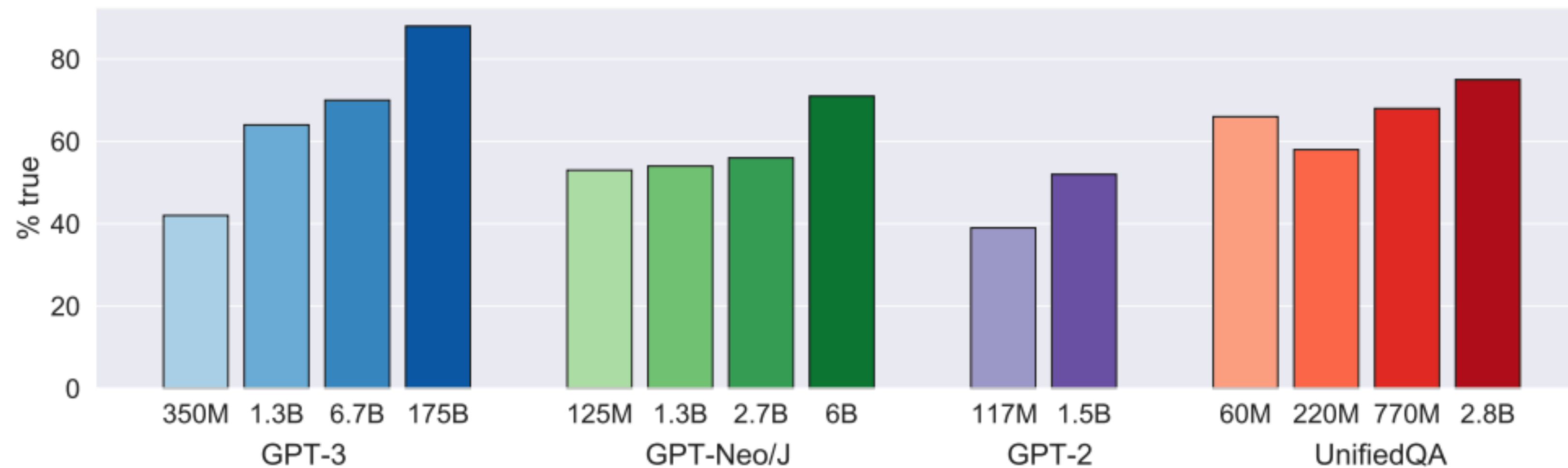


TruthfulQA

Average truthfulness on our benchmark



Average truthfulness on control trivia questions





Factuality

- ▶ Language models model distributions over text, not facts. There's no guarantee that what they generate is factual:
 - ▶ Language models are trained on the web. Widely-popularized falsehoods may be reproduced in language models
- ▶ A language model may not be able to store all rare facts, and as a result moderate probability is assigned to several options
- ▶ There are many proposed solutions to factuality. How do we evaluate them? How do we check facts "explicitly"?

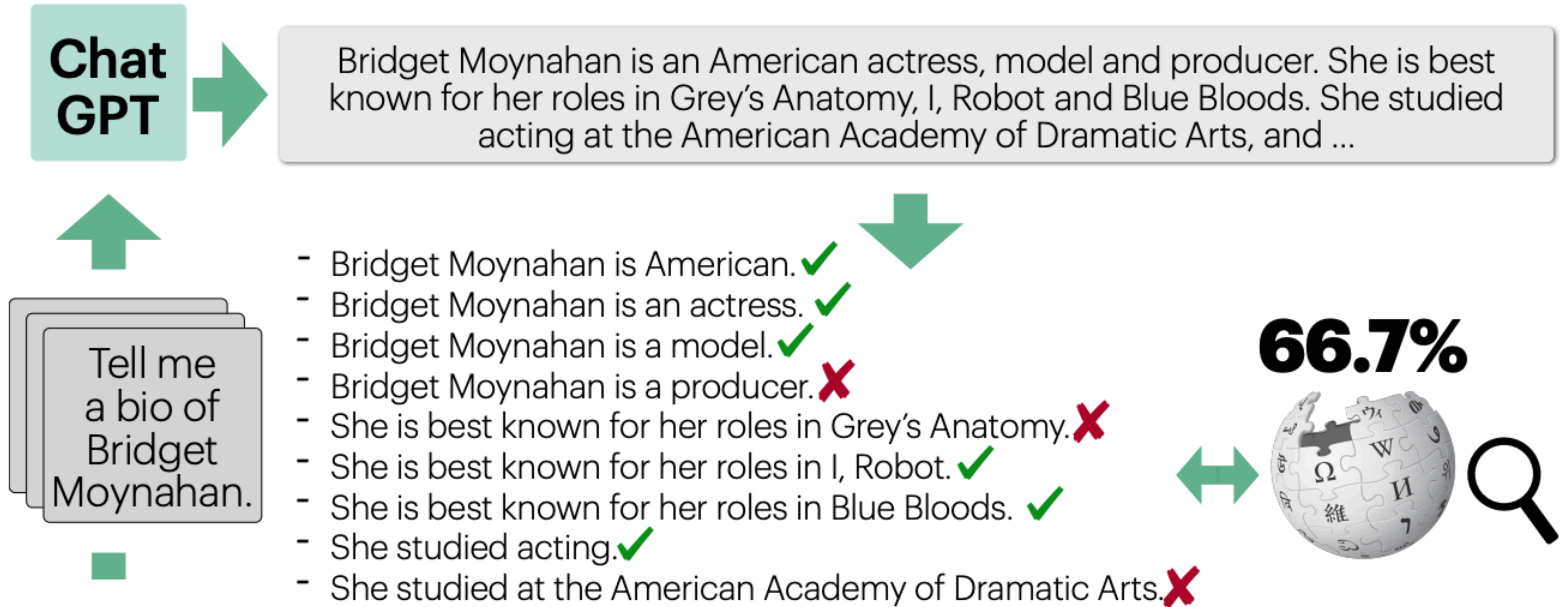


Grounding LM Generations

- ▶ Suppose we have text generated from an LM. We want to check it against a source document. What techniques have we seen so far that can do this?
- ▶ What steps are involved?
 1. Decide what text you are grounding in (may involve retrieval)
 2. Decompose your text into pieces of meaning to ground
 3. Check each piece
- ▶ For now, we'll assume the reference text/documents are given to us and not focus on step 1



Concrete Setting



- Dataset: ChatGPT-generated biographies of people. May contain errors, particularly when dealing with obscure people!



Step 2: Decomposition

- ▶ Simplest approach: each sentence needs to be grounded
- ▶ Can go deeper: think of sentences as expressing a collection of propositions
- ▶ Long history in frame semantics of defining these propositions. Many propositions anchor to verbs
- ▶ Recent work: extract propositions with LLMs

Original Sentence:

The main altar houses a 17th-century fresco of figures interacting with the framed 13th century icon of the Madonna (1638), painted by Mario Balassi.

GPT-3

- The main altar houses a 17th-century fresco.
- The fresco is of figures interacting with the framed 13th-century icon of the Madonna.
- The icon of the Madonna was painted by Mario Balassi in 1638.

Yixin Liu et al. (2023)

Ryo Kamoi et al. (2023)



Takeaways

- ▶ Chain-of-thought prompting (zero- and few-shot) can work well for tasks involving reasoning, especially mathematical reasoning and textual question answering with multiple steps
- ▶ Several things needed to improve them, such as self-consistency and the ability to use other resources like code execution or APIs
- ▶ Next time: building an LLM: pretraining, midtraining, instruction-tuning